Implement Backtracking search to solve the 8-Queens constraint satisfaction problem.

```
N = 8

board = [-1] * N



def is_safe(row, col):

    for previous_row in range(row):

        previous_col = board[previous_row]

        # Check same column
        if previous_col == col:
            return False

        # Check same diagonal
        if abs(previous_row - row) == abs(previous_col - col):
            return False

    return True



def solve(row):

    # All queens are placed
    if row == N:
        return True

    # Try every column
```

```
    for col in range(N):

        if is_safe(row, col):

            # Place queen
            board[row] = col

            # Solve for next row
            if solve(row + 1):
                return True

            # Backtracking
            board[row] = -1

    return False


# Solve the problem
if solve(0):

    print("Solution found:\n")

    for row in range(N):

        for col in range(N):

            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
```

```
        print()


else:
    print("No solution found")
#RUN
python eight_queens.py
```